Weapon-Target Assignment Problem

CSE 462: Algorithm Engineering Sessional
Checkpoint 1 Presentation

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Real-World Motivation: Iron Dome

Israel's Missile Defense System

Uses WTA algorithms to assign interceptor missiles to incoming rockets in real-time with 90% success rate.

- Incoming Rockets = **Targets**
- Interceptor Missiles = **Weapons**

→ Assignment = **WTA Solution**

Why WTA Matters

- Real-time decision making
- Limited interceptors vs multiple threats
- Maximize protection of high-value assets

Iron Dome: WTA in Action

Source: Israeli Defense Forces

Presentation Outline

Part 1: Theory

1. **Problem Definition**
   - Formal definition
   - Mathematical formulation
   - Classifications
2. **NP-Hardness Proofs**
   - Reduction from SUBSET PRODUCT
   - Reduction to Multiple Knapsack
3. **Algorithm Survey**
   - Exact, approximation, heuristic algorithms

Part 2: Implementation

4. **Selected Algorithms**
   - Maximum Marginal Return (MMR)
   - Ant Colony Optimization (ACO)
5. **Experimental Design**
   - Benchmark instances
   - Metrics and analysis plan
6. **Applications**
   - Military and civilian uses

Total Time: 15 minutes (6 sections $\times$ 2.5 minutes each)

What is the Weapon-Target Assignment Problem?

A fundamental combinatorial optimization problem in operations research

Problem Definition

Weapon-Target Assignment (WTA) Problem

The WTA problem involves the **strategic allocation of a set of weapons to a set of targets** to achieve a specific defensive or offensive objective.

Core Challenge: Stochastic Nature of Combat

Every engagement is modeled as a **probabilistic event** where a weapon has a specific likelihood of destroying its assigned target.

Computational Complexity

- The problem is mathematically proven to be **NP-complete**
- As the number of weapons and targets grows, complexity increases **exponentially**
- Finding an absolute optimal solution becomes extremely difficult for large instances

Classification: Asset-based vs. Target-based

Two Perspectives of WTA

The WTA problem can be viewed from two distinct perspectives depending on whether the objective prioritizes **destruction of enemy forces** or **protection of friendly resources**.

Target-based Perspective

Values assigned to each **enemy target** based on threat they pose.

- **Goal:** Maximize total damage inflicted on incoming targets
- **Math: Minimize** expected survival value of targets
- (Equivalent to maximizing damage)

Asset-based Perspective

Values assigned to **defended assets** (cities, bases) rather than targets.

- **Goal:** Maximize combined survival value of friendly assets
- **Requirement:** Map which targets attack which assets
- **Math: Maximize** expected value of surviving assets

Mathematical Representation: Parameters and Variables

Target-based Formulation – Parameters

- $W = \{1, \ldots, m\}$: The set of available **weapon types**
- $T = \{1, \ldots, n\}$: The set of **targets** to be engaged
- v_j: The **strategic military value** or importance of target j
- p_{ij}: The "**probability of kill**" – chance that one weapon of type i will destroy target j
- w_i: The **total quantity** of weapons of type i available

Decision Variable

$$x_{ij} \in \{0, 1, 2, \ldots\}$$

Integer variable representing the number of weapons of type i assigned to target j

Mathematical Representation: Objective Functions

Goal: Maximize damage (or equivalently, minimize survival)

While the goal is to maximize damage, the standard convention is to minimize remaining survival value.

1. Maximizing Total Expected Damage

$$\text{Max } Z = \sum_{j=1}^{n} v_j \left[1 - \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}} \right]$$

2. Minimizing Expected Survival Value

$$\text{Min } Z = \sum_{j=1}^{n} v_j \prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$$

Key Insight

$\prod_{i=1}^{m} (1 - p_{ij})^{x_{ij}}$ = **Probability that target j survives ALL assigned weapons**
The product captures the **combined effect** of multiple weapons attacking the same target.

Mathematical Representation: Constraints

Constraint 1: Resource Availability

$$\sum_{j=1}^{n} x_{ij} \leq w_i, \quad \forall i \in W$$

Meaning: Total weapons of type i assigned across all targets cannot exceed available quantity w_i

Constraint 2: Integrality

$$x_{ij} \in \{0, 1, 2, \dots\}$$

Meaning: We can only assign **whole numbers** of weapons (discrete assignments)

Complete Formulation Summary

Minimize $\sum_j v_j \prod_i (1 - p_{ij})^{x_{ij}}$ **s.t.** $\sum_j x_{ij} \leq w_i, \quad x_{ij} \in \mathbb{Z}^+$

Linear vs. Non-Linear Assignment

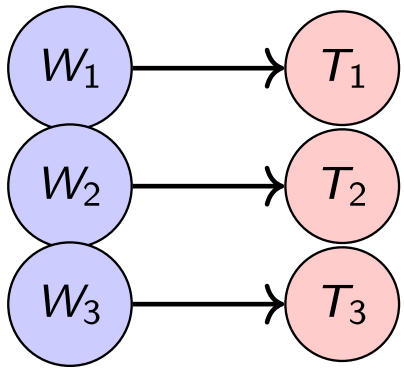
Linear Assignment Problem (LAP)

- **Structure:** One weapon assigned to exactly one target (**One-to-One**)
- **Linearity:** Objective function is linear
- Assignments are **independent**
- **Polynomial time solvable** (Hungarian algorithm)

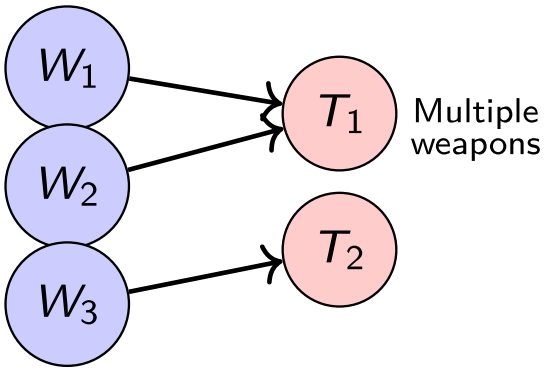
Non-Linear Assignment (Non-LAP)

- **Structure:** Multiple weapons can be assigned to single target (**Many-to-One**)
- **Non-Linearity:** Due to "**diminishing returns**"
- Additional weapons = **overkill** effect
- **NP-complete!**

LAP (One-to-One)



Non-LAP (Many-to-One)



Polynomial-Time Reductions

Proving WTA is NP-complete through formal reductions

WTA Decision Problem

Decision Version of WTA

Given:

- Survival probability $q_{i,j} = 1 - p_{i,j}$
- Threshold value k

Question: Does there exist an assignment $x_{i,j} \in \{0, 1\}$ such that:

$$D = \sum_{j=1}^{n} V_j \prod_{i=1}^{m} q_{i,j}^{x_{i,j}} \leq k$$

Subject to Constraints

$$\sum_{j=1}^{n} x_{i,j} \leq W_i \quad \text{for } i = 1, \ldots, m$$

$$x_{i,j} \in \{0, 1\}$$

where $x_{i,j} = 1$ if weapon i assigned to target j, and $x_{i,j} = 0$ otherwise.
D represents **total expected damage** (weighted survival probability of all targets)

Two Polynomial-Time Reductions

Our Approach

We demonstrate two polynomial-time reductions:

Reduction 1

SUBSET PRODUCT $\leq_p$ WTA

$\Rightarrow$ Proves **WTA is NP-hard**

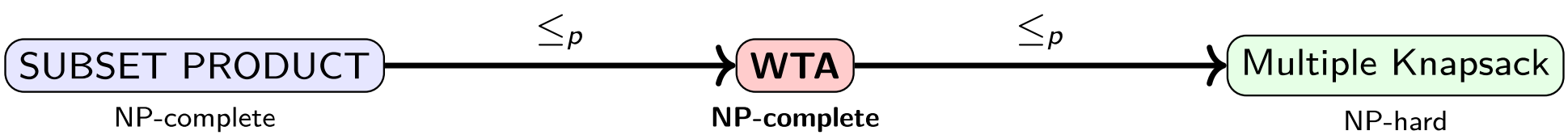
(Reduce FROM a known NP-complete problem TO WTA)

Reduction 2

WTA $\leq_p$ Multiple Knapsack

$\Rightarrow$ Shows WTA reduces to another known NP-hard problem

(Shows connection between problems)



Reduction 1: SUBSET PRODUCT $\leq_p$ WTA

SUBSET PRODUCT Problem (NP-complete)

Given: Set $A = \{a_1, a_2, \ldots, a_m\}$ and target product b
Find: Subset $X \subseteq A$ where $\prod_{x \in X} a_x = b$

WTA Instance Construction

Let $P = \prod_{i=1}^{m} a_i$. Construct WTA with:

- **Targets:** 2 targets with values $V_1 = V_2 = P$
- **Weapons:** $m + 2$ weapons: $W_1, W_2, \ldots, W_m, W_\alpha, W_\beta$
- **Survival probabilities:**
  - $q_{i,1} = q_{i,2} = \frac{1}{a_i}$ for $i = 1, \ldots, m$
  - $q_{\alpha,1} = q_{\alpha,2} = \frac{b}{2P}$
  - $q_{\beta,1} = q_{\beta,2} = \frac{1}{2b}$
- **Threshold:** $k = 1$ (we ask if $D \leq 1$)

Time Complexity

Constructs WTA with 2 targets and $m + 2$ weapons in $O(m)$ time – **Polynomial!**

Reduction 1: Proof of Correctness

Forward Direction ($\Rightarrow$): SUBSET PRODUCT has solution $\Rightarrow$ WTA has solution

Suppose $X \subseteq \{1, \ldots, m\}$ with $\prod_{x \in X} a_x = b$. Let $Y = \{1, \ldots, m\} \setminus X$.
Then:

$$\prod_{i \in X} \frac{1}{a_i} = \frac{1}{b}, \quad \prod_{y \in Y} \frac{1}{a_y} = \frac{b}{P}$$

Assignment: $\{W_x : x \in X\} \cup \{W_\alpha\} \to T_1$ and $\{W_y : y \in Y\} \cup \{W_\beta\} \to T_2$
Total damage:

$$\begin{aligned} D &= V_1 \cdot \frac{b}{2P} \cdot \frac{1}{b} + V_2 \cdot \frac{1}{2b} \cdot \frac{b}{P} \\ &= P \cdot \frac{b}{2P} \cdot \frac{1}{b} + P \cdot \frac{1}{2b} \cdot \frac{b}{P} = \frac{1}{2} + \frac{1}{2} = 1 \leq k \quad \checkmark \end{aligned}$$

Backward Direction ($\Leftarrow$): WTA has solution $\Rightarrow$ SUBSET PRODUCT has solution

Suppose WTA has solution with $D \leq 1$. Let X', Y' be weapons assigned to T_1, T_2.
Claim: W_α and W_β cannot both target the same target.
Proof: If both assigned to T_1 (symmetric for T_2):

$$D \geq P \cdot \frac{1}{2b} \cdot \frac{b}{2P} \cdot \frac{1}{\prod_{x \in X' \setminus \{\alpha, \beta\}} a_x} + \frac{P}{\prod_{y \in Y'} a_y} \geq \frac{1}{4} + 1 > 1 \quad \Rightarrow\Leftarrow$$

Reduction 1: Backward Direction (Continued)

From the Assignment

With $W_\alpha \to T_1$ and $W_\beta \to T_2$:

$$D = P \cdot \frac{b}{2P} \prod_{x \in X} \frac{1}{a_x} + P \cdot \frac{1}{2b} \prod_{y \in Y} \frac{1}{a_y} \leq 1$$

Algebraic Derivation

Let $z = \prod_{x \in X} a_x$. Then:

$$\frac{b}{2z} + \frac{z}{2b} \leq 1$$

Multiply by $2bz > 0$:

$$b^2 + z^2 \leq 2bz \quad \Rightarrow \quad (b - z)^2 \leq 0$$

Since $(b - z)^2 \geq 0$, we must have $(b - z)^2 = 0$, so:

$$b = z = \prod_{x \in X} a_x$$

Conclusion

X is a valid solution to SUBSET PRODUCT! ✓

Time Complexity: $O(m)$ – Polynomial reduction complete

Reduction 2: WTA $\leq_p$ Multiple Knapsack

Multiple Knapsack Construction

Given WTA with m targets, n weapons, probabilities $\{p_{i,j}\}$:

- **Knapsacks:** m knapsacks (one per target)
- **Capacity:** $C_j = \infty$ (or sufficiently large)
- **Items:** n items (one per weapon type)
- **Profit:** $v_{ij} = -\log(1 - p_{i,j}) = -\log(q_{i,j})$
- **Weight:** $w_{ij} = 0$ (or 1 if weapon limits exist)

Objective Transformation (Key Insight)

For single target j, minimizing $\prod q_{i,j}^{x_{i,j}}$ equals minimizing:

$$\log\left(\prod_{i=1}^{n} q_{i,j}^{x_{i,j}}\right) = \sum_{i=1}^{n} x_{i,j} \log(q_{i,j})$$

Equivalent to **maximizing**:

$$-\sum_{i=1}^{n} x_{i,j} \log(q_{i,j}) = \sum_{i=1}^{n} x_{i,j} \cdot v_{ij}$$

This is **exactly the Multiple Knapsack objective!**

Reduction 2: Constraint Mapping & Conclusion

Constraint Mapping

- **WTA constraint:** $\sum_{j=1}^{m} x_{i,j} \leq W_i$ (weapon type i used at most W_i times)
- **Maps to:** $\sum_{j=1}^{m} y_{ij} \leq 1$ (each item in at most one knapsack) when $W_i = 1$
- For general W_i: Create W_i copies of item i

Equivalence

Assignment $x_{i,j}$ optimal for WTA $\Leftrightarrow$ $y_{ij} = x_{i,j}$ optimal for Multiple Knapsack
Logarithmic transformation preserves ordering of solutions!
Time: $O(nm)$ to compute all $v_{ij} = -\log(1 - p_{i,j})$

Final Conclusion

WTA is in NP (solution verifiable in polynomial time) + **WTA is NP-hard**
$\Rightarrow$ **The Weapon-Target Assignment Problem is NP-complete!**